Aman Panchal

+91-9350078968 | amanpanchal3110@gmail.com | linkedin.com/in/aman-panchal | github.com/AmanPanchal3110

Summary

Data Engineer with hands-on experience designing and deploying end-to-end ETL/ELT data pipelines and data models using Apache Kafka, Apache Spark, Apache Airflow, dbt Core, Snowflake, and CDC with Debezium. Proficient in Medallion Architecture (Bronze/Silver/Gold), data warehousing, real-time stream processing, batch processing, and data quality enforcement across AWS S3, PostgreSQL, and Delta Lake. Experienced in containerising production-grade data stacks with Docker and delivering business-ready analytics via Metabase and Power BI.

Education

Indian Institute of Information Technology, Nagpur *2023 – 2027*
B.Tech in Electronics and Communication Engineering (IoT)

Technical Skills

Languages: Python, SQL, C, C++
Data Engineering: Apache Kafka, Apache Spark, PySpark, Apache Airflow, dbt, CDC, Debezium, ETL/ELT Pipelines, Data Modeling, Data Warehousing, Medallion Architecture, Stream & Batch Processing, Workflow Orchestration, Data Validation, Data Quality, Fault-Tolerant Pipelines, Real-Time Data Processing
Cloud & Storage: AWS S3, Snowflake, PostgreSQL, BigQuery, Delta Lake, Parquet, External Stages, REST APIs
Analysis & Visualization: Pandas, NumPy, SQL Query Optimization, Metabase, Power BI, Exploratory Data Analysis, Web Scraping
Tools & DevOps: Docker, Docker Compose, Git, GitHub, Linux, DBeaver

Projects

Rapido CDC Real-Time Data Pipeline *GitHub*
- **Stack:** Python, PostgreSQL, Debezium, Apache Kafka (KRaft), Spark 3.5 Structured Streaming, Delta Lake, AWS S3, Airflow 3, Snowflake, dbt Core, Docker Compose
- Built real-time CDC pipeline capturing INSERT/UPDATE/DELETE events from PostgreSQL via Debezium WAL log reading, streamed through 6 Kafka partitions into a 24/7 Spark Structured Streaming Bronze layer on AWS S3 processing ~60–70 events/sec
- Orchestrated Airflow DAG (every 30 min) starting at Silver: incremental Delta Lake CDF transformation with MERGE and OPTIMIZE/ZORDER to avoid full table scans, then loaded curated data into Snowflake STAGING via Spark
- Modelled Snowflake MARTS layer in dbt (dimensions, facts, aggregates, one-big-table) with data quality tests (uniqueness, referential integrity, accepted ranges) triggered via `dbt run` → `dbt test` → `dbt docs` at the end of the DAG

E-Commerce Order Intelligence Platform *GitHub*
- **Stack:** Python, Apache Kafka (KRaft), Spark 3.5 Structured Streaming, AWS S3, Airflow 3, Snowflake, dbt, Metabase, Docker Compose
- Simulated production-grade e-commerce events (orders, payments, returns) via Kafka and built Spark Structured Streaming pipeline writing Delta/Parquet files to AWS S3 across 4 structured paths
- Orchestrated end-to-end Airflow 3 DAG for S3-to-Snowflake ingestion via CopyFromExternalStage with dbt Bronze → Silver → Gold incremental transformations; implemented data validation checks and quality controls for reliable data delivery
- Modelled 5 Gold-layer analytics tables connected to Metabase dashboards for business insights; containerised the full stack using Docker Compose with custom network and health checks

Crypto Market Data Pipeline *GitHub*
- **Stack:** Python, Apache Airflow, AWS S3, PostgreSQL, dbt, Docker, PyArrow
- Ingested live CoinGecko API data as date-partitioned Parquet files into AWS S3; orchestrated extraction, transformation, S3 upload, and PostgreSQL load via Airflow TaskFlow API with retry and failure handling
- Implemented Medallion Architecture in dbt (Bronze → Silver → Gold) with incremental models, dimension/fact tables, and analytical marts for market trend analysis; applied data validation checks for schema consistency and missing data handling

Achievements

- Solved 120+ LeetCode problems (data structures and algorithms) and 50+ SQL problems on DataLemur (window functions, CTEs, joins, aggregations)
- Secured top 10 position in the Analytica Data Science Hackathon among 100+ competing teams; familiar with Agile development practices including sprint planning, iterative delivery, and collaborative problem-solving